

LECTURE -17

1. Molecular Genetics of Antibody Diversity

This section details the structure of the antibody (B cell receptor) and the genetic mechanism used to generate its immense diversity.

Antibody Structure and Components¹¹¹¹¹¹¹¹¹¹¹¹¹¹¹¹¹¹¹¹

- A B cell receptor (or secreted antibody) is made of **two identical heavy chains** and **two identical light chains**².
- The chains are held together by **disulfide bridges**³.
- Each receptor has **two identical Antigen-binding sites**⁴⁴.
- It is divided into two regions⁵⁵⁵⁵:
 - **Variable (V) regions**: Located at the tips of the 'Y' shape, these regions form the **Antigen-binding site**⁶⁶⁶⁶. They vary extensively in sequence⁷.
 - **Constant (C) regions**: Determine the antibody class (e.g., IgM, IgG) and are located closer to the cell membrane⁸.
- **Chains and Domains** (as shown in the immunoglobulin G (IgG) model)⁹⁹⁹⁹⁹⁹⁹⁹⁹⁹⁹⁹⁹⁹⁹⁹:
 - **Heavy chain** has Variable (V_H) and Constant (C_H1 , C_H2 , C_H3) domains¹⁰¹⁰¹⁰¹⁰¹⁰¹⁰¹⁰¹⁰¹⁰¹⁰¹⁰¹⁰¹⁰¹⁰¹⁰.
 - **Light chain** has Variable (V_L) and Constant (C_L) domains¹¹¹¹¹¹¹¹¹¹¹¹¹¹¹¹¹¹.
 - The **hinge** region provides flexibility¹².
- **Complementarity Determining Regions (CDRs)**: The antigen-binding site is formed by **CDR1, CDR2, and CDR3** loops within the V regions of both chains. **CDR3** is generated during recombination and is the most variable, contributing most to antigen specificity¹³¹³¹³¹³.

The Genetic Basis of Diversity (V(D)J Recombination)

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- The genes coding for the V, D, and J regions of B and T cell receptors **undergo recombination**¹⁵.
- This process, called **somatic recombination**, occurs in **hematopoietic stem cells of the bone marrow** during B cell development¹⁶¹⁶¹⁶¹⁶.
- **Heavy Chain Variable Region** is made of **3 segments**: **V** (Variable), **D** (Diversity), and **J** (Joining)¹⁷.
- **Light Chain Variable Region** is made of **2 segments**: **V** and **J**¹⁸. The "**D**" region is **only on the variable segment of the heavy chain**¹⁹.
- **Loci and Gene Segments**²⁰: The genes are clustered in **three loci** in the human genome²¹:
 - **IGH (Heavy-chain locus)** on chromosome 14, containing ≈ 50 { V }\$, 23 { D }\$, 6 { J }\$, and 9 { C }\$ segments²².
 - **IGK (κ light-chain locus)** on chromosome 2, containing ≈ 40 { V }\$, 5 { J }\$, and 1 { C }\$ segment²³.
 - **IGL (λ light-chain locus)** on chromosome 22, containing ≈ 32 { V }\$, 5 { J }\$, and 5 { C }\$ segments²⁴.
- **Diversity Mechanisms**²⁵: Mechanisms like **combinatorial V-(D)-J diversity**, **junctional diversity** (random addition/deletion of nucleotides at junctions), and **somatic hypermutations** lead to an estimated potential repertoire of 10^{12} different immunoglobulins²⁶.

2. T Cell Subsets and Immune Tolerance

This section addresses the lineage of immune cells, the differentiation of T helper cells, and the mechanisms preventing autoimmunity.

Haematopoiesis (Review)²⁷²⁷²⁷²⁷²⁷

- All immune cells come from **Hematopoietic stem/Progenitor cells**²⁸.

- Lymphoid Lineage** (via Common Lymphoid progenitor cells and Lymphoblast)
 produces **Lymphocytes (B and T Lymphocytes)** and **NK cells**

T Helper Cell Differentiation

When a $\text{Naive~CD4}^+\text{~T~cell}$ is presented with an antigen by an $\text{Antigen~Presenting~Cell}$ (Dendritic cell) via the MHC-II/TCR complex, it undergoes **Clonal Expansion** and differentiates into specialized T~helper~subsets based on specific cytokines and transcription factors:

T Helper Subset	Cytokines/Transcription Factors	Primary Function
Th1	IFN- γ , IL-2, T-bet	Cell Mediated Immunity (Responses to intracellular pathogens like viruses)
Th2	IL-4, IL-13, Gata-3	Humoral Immunity (Responses to extracellular pathogens like helminths) and Allergy
Th17	IL-6, TGF- β , ROR γ	Mucosal Immunity (Responses to extracellular/intracellular bacteria and fungi)

Treg	IL-10, TGF- β , FoxP3 ³⁸		Immune Suppression (Maintains tolerance and mucosal homeostasis) ³⁹
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Immune Tolerance⁴⁰⁴⁰⁴⁰⁴⁰⁴⁰

Tolerance is the mechanism that prevents the immune system from attacking the host's own tissues (autoimmunity).

- **Central Tolerance:** Occurs in the **Thymus or Bone marrow** with **Immature B or T cells**⁴¹. Cells with receptors that recognize self-antigens are eliminated by **Apoptosis** (programmed cell death)⁴².
- **Peripheral Tolerance:** Occurs in **Secondary lymphoid tissues** with **Mature B or T cells**⁴³⁴³⁴³⁴³⁴³⁴³⁴³⁴³. If self-recognizing cells escape central tolerance, they can be suppressed by $\mathbf{Tregs}$ or deactivated through:
 - **Anergy** (functional inactivation)⁴⁴.
 - **Activation Induced Cell Death (AICD)**⁴⁵.
- **Significance:** The discovery of $\mathbf{Tregs}$ (Regulatory T cells) and peripheral tolerance (attributed to scientists like Shimon Sakaguchi⁴⁶) is **key to developing therapies for cancer and autoimmune diseases**⁴⁷.

3. DNA Recombination and Chromosome Structure

This section defines DNA recombination and connects it to the structure and division of chromosomes.

DNA Recombination⁴⁸⁴⁸⁴⁸⁴⁸⁴⁸⁴⁸⁴⁸⁴⁸

- **Definition:** Involves the **exchange of genetic material**⁴⁹⁴⁹⁴⁹⁴⁹.
- **Location:** Can occur between different regions of the same chromosome (e.g., generation of antibody diversity)⁵⁰⁵⁰⁵⁰⁵⁰ or between **multiple chromosomes**⁵¹⁵¹⁵¹⁵¹.

- **Types:**
 - **Meiosis:** Recombination between **non-sister (homologous) chromosomes** results in **genetic diversity**⁵². This occurs during **Prophase of meiosis I**⁵³.
 - **Mitosis:** Recombination between **sister chromatids** is usually used to **repair DNA** (restoring lost information due to damage)⁵⁴. This occurs during the **S or G2 phase of the cell cycle**⁵⁵.
- **Holliday Junction:** The exchange of DNA segments results in the formation of a four-way junction known as a "**Holliday Junction**"⁵⁶⁵⁶⁵⁶⁵⁶. These junctions must be **resolved** to separate the DNA segments, and the **type of resolution results in different end products** for the exchange event⁵⁷⁵⁷⁵⁷⁵⁷.

Chromosome Structure⁵⁸⁵⁸⁵⁸⁵⁸

- **Homologous Chromosomes:** A pair of chromosomes that are the same⁵⁹⁵⁹⁵⁹⁵⁹.
- **Sister Chromatids:** One half of a duplicated chromosome⁶⁰⁶⁰⁶⁰⁶⁰. Sister chromatids come into existence in the **S-phase of mitosis** (or the cell cycle)⁶¹.
- **Key Regions:**
 - **Centromere:** The constriction point where the two sister chromatids are joined⁶².
 - **Telomere:** The ends of the chromosome arms⁶³.
 - **Short Arm (\$\text{P}\$)** and **Long Arm (\$\text{Q}\$)**⁶⁴⁶⁴⁶⁴⁶⁴.

Mitosis and Meiosis Questions

This set of questions relates chromosome structure to cell division processes:

- **What does mitosis separate?** $\rightarrow$ **Sister Chromatids** (during Anaphase).
- **What does meiosis separate?** $\rightarrow$ **Homologous Chromosomes** (Meiosis I) and **Sister Chromatids** (Meiosis II).
- **What does the mature sperm/egg contain?** $\rightarrow$ **Chromosomes** (specifically, a haploid set of unduplicated chromosomes, one chromatid each).
- **What does the one-cell embryo contain?** $\rightarrow$ **Chromosomes** (a diploid set of unduplicated chromosomes).
- **What does the interphase cell contain?** $\rightarrow$ **Chromosomes** (unduplicated in G1, duplicated in G2—meaning a mix of single and double chromatid chromosomes).

Haploid (n)

A **haploid** cell contains **one set of chromosomes**¹.

- **Symbol:** Represented as **n** .
- **Source:** Haploid cells are typically **gametes** (sex cells), such as the **mature sperm**² and the **mature egg**³.
- **Function:** Gametes are produced through **meiosis**⁴ and are designed to fuse during fertilization to create a new organism. Each chromosome is **unduplicated** (consisting of a single chromatid)⁵.

Diploid (2n)

A **diploid** cell contains **two sets of chromosomes**⁶.

- **Symbol:** Represented as **$2n$** .
- **Source:** This includes **somatic cells** (body cells) and the **one-cell embryo** (zygote)⁷.
- **Sets:** One set of chromosomes is inherited from each parent, making up **homologous chromosomes** (a pair of the same chromosomes)⁸.
- **Cell State:**
 - During **interphase**⁹ (G1 phase), a diploid cell contains two sets of unduplicated chromosomes.
 - After **S-phase** (DNA synthesis), the chromosomes are duplicated, meaning each chromosome consists of two identical **sister chromatids**¹⁰¹⁰¹⁰¹⁰.
- **Function:** Diploid cells divide through **mitosis**¹¹ to produce two identical diploid daughter cells. Mitosis separates the **sister chromatids**¹².

Central Tolerance

Central tolerance is the primary mechanism for preventing autoimmunity and occurs early in the development of immune cells.

- **Location** 📍: Occurs in the **primary lymphoid organs** where immune cells mature: the **Thymus** (for T cells) and the **Bone marrow** (for B cells)².

- **Target** ★: **Immature B or T cells** that have just generated their unique antigen receptors³.
- **Mechanism** ✖: During maturation, immune cells are tested against the body's own proteins (**self-antigens**). If a cell's receptor strongly recognizes or binds to a self-antigen, that cell is flagged as potentially dangerous.
- **Outcome** 💀: The cell is eliminated by **Apoptosis** (programmed cell death)⁴. This weeds out the vast majority of self-reactive cells before they enter circulation.

Peripheral Tolerance

Peripheral tolerance acts as a crucial **backup system** in the event that self-reactive mature immune cells escape central tolerance.

- **Location** 🌿: Occurs in the **secondary lymphoid tissues** (like the spleen, lymph nodes) and other body tissues⁵.
- **Target** 🛡️: **Mature B or T cells** that have already been released into the body⁶.
- **Mechanism** 🔴: Self-reactive cells that encounter a self-antigen in the periphery without the proper co-stimulatory signals are either deactivated or suppressed. The mechanisms include:
 - **Anergy**: The T or B cell becomes functionally inactivated, meaning it cannot respond to future stimulation⁷.
 - **Activation Induced Cell Death (AICD)**: The cell is triggered to die⁸.
 - **Treg Suppression: Regulatory T cells (Treg)** actively suppress the activity of self-reactive Effector T cells and B cells⁹.
- **Significance** 🔑: The discovery of Tregs and peripheral tolerance is a key area of research, as this knowledge is fundamental to **developing therapies for cancer** (where tolerance must be broken) and **autoimmune diseases** (where tolerance must be restored)¹⁰.

A **Treg** is a **Regulatory T cell**¹, which is a critical subset of CD4^+ T helper cells specialized in **immune suppression** and maintaining **self-tolerance**².

Tregs are essential for preventing the immune system from attacking the body's own tissues (autoimmunity) and for balancing the intensity of immune responses³³³.

Key Characteristics of Tregs

- **Differentiation:** Tregs develop from a $\text{Naive CD4}^+ \text{ T cell}$ when it is activated in the presence of the cytokine **TGF- β** ⁴.
- **Master Regulator:** The differentiation and function of a Treg are controlled by the master transcription factor **FoxP3**⁵.
- **Primary Function: Immune Suppression**⁶. They actively inhibit the activity of other T cells and B cells that are either excessively reacting to a pathogen or mistakenly targeting host tissue⁷.
- **Key Cytokines:** Tregs secrete immunosuppressive cytokines, primarily **IL-10** and **TGF- β** ⁸.
- **Role in Tolerance:** They are crucial for **Peripheral tolerance**, acting as the main mechanism to suppress self-reactive T cells that have escaped the initial central tolerance filtering in the thymus⁹⁹⁹⁹.
- **Role in Health:** They play a vital role in the **Maintenance of mucosal homeostasis** (balancing the immune response in the gut and other mucosal surfaces)¹⁰.

The discovery of Tregs and their role in peripheral tolerance is significant, opening avenues for developing new therapies for **cancer** (by inhibiting Tregs to boost anti-tumor immunity) and **autoimmune diseases** (by enhancing Treg function)¹¹.

A **Holliday Junction** is a crucial intermediate structure formed during **DNA recombination**, where two double-stranded DNA molecules exchange genetic material¹¹.

Detailed Explanation

- **Structure** : A Holliday Junction is a cross-shaped structure formed when two homologous DNA helices are linked together by the exchange of two single strands²². It's a key intermediate in many forms of genetic exchange.
- **Formation** : It forms when single strands from two separate DNA molecules break and then cross over to pair with the complementary strands of the other molecule. This results in a four-way junction between the DNA segments³³³³.
- **Resolution** : After the exchange occurs, this junction must be **resolved** (cut) by specialized enzymes to separate the individual DNA segments. The way the Holliday Junction is resolved determines the final type of end result for the exchange event⁴⁴⁴⁴⁴⁴⁴⁴⁴⁴⁴⁴⁴⁴⁴⁴⁴⁴⁴⁴.
- **Biological Roles** :
 - **Genetic Diversity (Meiosis)**: During meiosis, recombination between homologous chromosomes creates genetic diversity, and the Holliday Junction is a required step⁵⁵⁵⁵.
 - **DNA Repair (Mitosis)**: During mitosis, recombination between sister chromatids can be used to repair DNA damage and restore lost information⁶⁶⁶⁶.
 - **Antibody Diversity**: It is also involved in the exchange of genetic material between different regions of the same chromosome, such as in the generation of antibody diversity⁷⁷⁷⁷.